

CSCE 46103/56103 Artificial Intelligence – Fall 2026

Heuristic Design: Admissibility and Consistency

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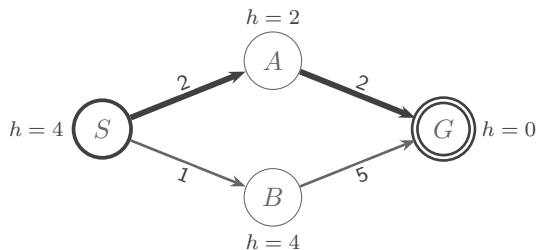
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Today's Topics

1. **A* Search Revisit:** selecting nodes using $f(n) = g(n) + h(n)$
2. **Heuristic Guarantees:** admissibility, consistency, and reopening
3. **Heuristic Design:** constructing and choosing effective heuristics
4. **Local Search:** hill climbing, escape strategies, population methods, and a challenge

A* Revisit: Choose by $g + h$, Not by g Alone

Graph and heuristic values



A* frontier trace: (g, h, f)

Step	Frontier	Select
Start	$S : (0, 4, 4)$	S
After S	$A : (2, 2, 4); B : (1, 4, 5)$	A
After A	$G : (4, 0, 4); B : (1, 4, 5)$	G

Result: $S \rightarrow A \rightarrow G$, with total cost 4.

True remaining costs: $h^*(S, A, B, G) = (4, 2, 5, 0)$

B has the smaller g , but A has the smaller f :
A* selects by **estimated total cost**.

Safe and Useful Heuristics

Strength

Would you prefer $h = 0$
or a larger estimate?

Safety

How large can $h(n)$
become before A^*
loses its guarantee?

Stability

Can every estimate be
optimistic, yet A^* still
need to undo work?

A good heuristic is a **lower bound**, a **stable guide**,
and **cheap enough to use**.

Admissibility and Consistency

$h^*(n)$: true cheapest cost from n to a goal

Admissible: test every node

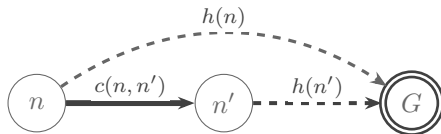
$$0 \leq h(n) \leq h^*(n).$$

The estimate is optimistic: it never exceeds the true cheapest remaining cost.

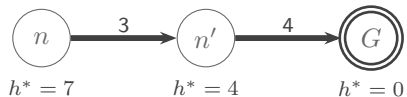
Consistent: test every edge

$$h(n) \leq c(n, n') + h(n').$$

The estimate cannot fall by more than the cost paid across one transition.



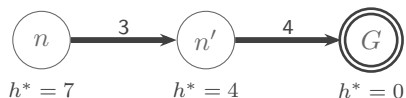
Easy Classification



	$h(n)$	$h(n')$	$h(G)$	Classification
A	6	4	0	
B	7	2	0	
C	8	4	0	

Classify each candidate before the reveal.

Easy Classification



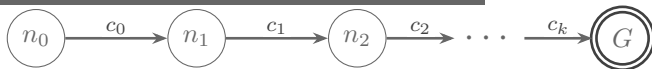
Classify each candidate before the reveal.

	$h(n)$	$h(n')$	$h(G)$	Classification
A	6	4	0	admissible and consistent
B	7	2	0	admissible; inconsistent because $7 > 3 + 2$
C	8	4	0	inadmissible because $8 > h^*(n)$

The distinction

Admissibility compares with an unknown global truth.
Consistency can be checked locally on known transitions.

Consistency-to-Admissibility Proof



Setup. Fix any goal path $P = (n_0, n_1, \dots, n_{k+1} = G)$ and write $c_i = c(n_i, n_{i+1})$. For every edge, consistency gives $h(n_i) \leq c_i + h(n_{i+1})$; at a goal, $h(G) = 0$.

$$\begin{aligned} h(n_0) &\leq c_0 + h(n_1) \\ &\leq c_0 + c_1 + h(n_2) \\ &\leq \dots \leq \sum_{i=0}^k c_i + h(G) \end{aligned}$$

consistency on the first edge
substitute the next edge bound

$$= \sum_{i=0}^k c_i = \text{cost}(P)$$

repeat along P

because $h(G) = 0$.

Consistency implies admissibility

$$h(n_0) \leq \min_{P: n_0 \rightsquigarrow G} \text{cost}(P) = h^*(n_0).$$

Converse fails: $(h(n), h(n')) = (7, 2)$ is admissible on a cost-3 edge, but $7 > 3 + 2$.

$$\begin{aligned} f(n') &= g(n') + h(n') \\ &= g(n) + c(n, n') + h(n') \\ &\geq g(n) + h(n) \\ &= f(n). \end{aligned}$$

What this buys A*

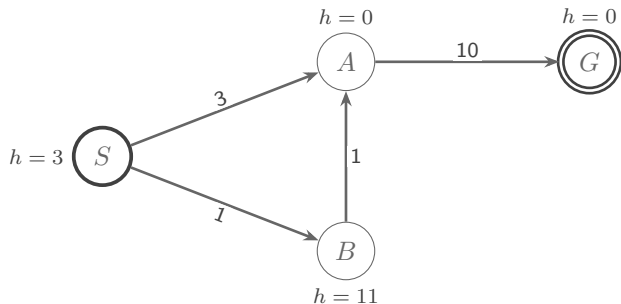
When a state is selected at minimum f , no later path can reveal a cheaper route that requires reopening that expanded state.

What it does not buy

A state still waiting on the frontier may receive a lower- g replacement before it is selected.

Consistency removes **expanded-state reopens**, not all best- g updates.

Classify This Heuristic



True remaining costs

$$h^*(A) = ?, \quad h^*(B) = ?, \\ h^*(S) = ?.$$

Are all estimates admissible?

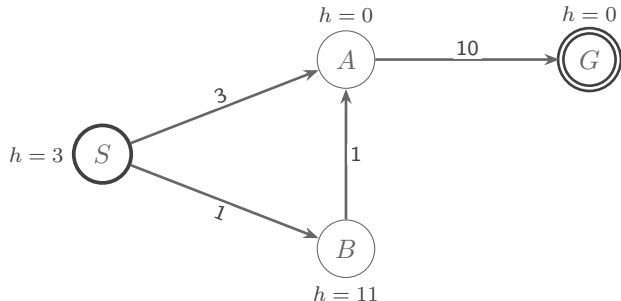
?

Are all edges consistent?

?

Compute h^* , decide admissibility, and test consistency.

Classify This Heuristic



True remaining costs

$$h^*(A) = 10, \quad h^*(B) = 11, \\ h^*(S) = 12.$$

All estimates are admissible

$$3 \leq 12, \quad 0 \leq 10, \quad 11 \leq 11.$$

No—one edge is inconsistent

$$h(B) = 11 > 1 + h(A) = 1.$$

Question: which state does A^* expand before it discovers the cheapest path to that state?

Reopening Trace

Step	Selected	Frontier after processing	Consequence
1	$S : (g, f) = (0, 3)$	$A : (3, 3), B : (1, 12)$	low $h(A)$ makes A look urgent
2	$A : (3, 3)$	$B : (1, 12), G : (13, 13)$	A is now expanded on a nonoptimal path
3	$B : (1, 12)$	$A : (2, 2), G : (13, 13)$	cheaper path forces reopening of A
4	$A : (2, 2)$	$G : (12, 12)$	goal record improves from 13 to 12
5	$G : (12, 12)$	—	return the optimal cost 12

With best- g reopening

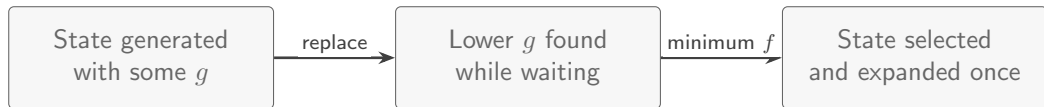
Returned path: $S \rightarrow B \rightarrow A \rightarrow G$, cost **12**.

If expanded A is never reopened

The cheaper parent chain cannot reach G ; the remaining goal costs **13**.

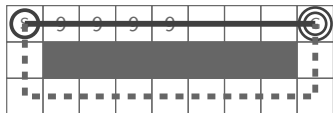
Consistency and Duplicate Updates

	Admissible, inconsistent h	Consistent perfect h^*
Values at (S, A, B, G)	$(3, 0, 11, 0)$	$(12, 10, 11, 0)$
After expanding S	select A at $f = 3$	select B at $f = 12$
Cheaper route to A	found after A expanded	replaces $A : g = 3$ while A waits
Expanded-state reopen	required	unnecessary
Returned cost	12 with reopening	12



The guarantee attaches to **selection/expansion**, not first generation.

Weighted Manhattan Certificate



Direct route: cost $4(9) + 4(1) = 40$

Lower detour: cost **12**

Supplied heuristic

$$h(n) = d_1(n, G)c_{\min}.$$

At the start, $h(S) = 8(1) = 8 \leq 12$.

Edgewise certificate

For every move $n \rightarrow n'$,

$$d_1(n, G) \leq 1 + d_1(n', G)$$

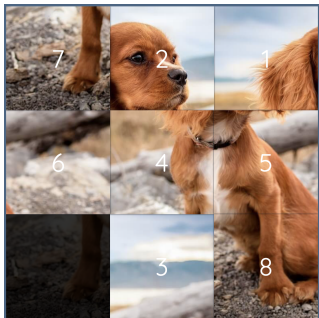
and $c(n, n') \geq c_{\min}$. Therefore,

$$h(n) \leq c(n, n') + h(n').$$

Safe does not mean exact: Manhattan ignores walls and expensive terrain.

The 8-Puzzle as a Search Problem

Start state



slide
adjacent
tiles



Try it live

Completed image



Search formulation

State: tiles 1–8 and one blank on a 3×3 board. **Action:** slide one orthogonally adjacent tile into the blank. **Goal:** place every numbered tile in its target square.

Design an 8-Puzzle Heuristic

State n

7	2	4
5		6
8	3	1

Goal

	1	2
3	4	5
6	7	8

Invent candidate heuristics

1. What work must *every* solution perform?
2. Turn that work into a cheap numerical estimate $h(n)$.
3. Can you invent a second estimate that is more informative?

Stress-test each idea

Does $h(\text{goal}) = 0$? Could $h(n)$ ever exceed the true minimum number of moves?

Be ready to share **one formula** and **one reason it should not overestimate**.

Potential Methods

A heuristic can be any function—but not every function is **safe** or **useful**.

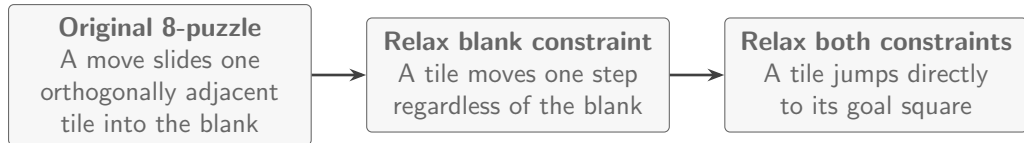
- ▶ Count the misplaced numbered tiles.
- ▶ Sum each tile's grid distance to its goal.
- ▶ Double the grid-distance estimate.
- ▶ Sum the numbers printed on misplaced tiles.
- ▶ Choose a random number.
- ▶ Solve the puzzle optimally and use the exact remaining cost.

Test every proposal

$$h(\text{goal}) = 0, \quad h(n) \leq h^*(n), \quad \text{cheap to compute?}$$

Which are safe? Which are useful? Which fail immediately?

Heuristics from Relaxations



Original optimum

May require interactions among tiles and the blank.

Manhattan h_2

Exact cost in the adjacent-but-occupancy-free relaxation.

Misplaced tiles h_1

Exact cost when each misplaced tile may jump into place.

Removing restrictions can only shorten an optimal solution: $h_{\text{relaxed}}(n) \leq h^*(n)$.

Misplaced Tiles vs. Manhattan

Easy instance

Start			Goal		
1	4	2		1	2
6		5	3	4	5
7	3	8	6	7	8

$$h_1 = 5, \quad h_2 = 6, \quad h^* = 6.$$

Five tiles are misplaced; tile 3 alone is two grid steps from its goal.

Hard instance

Start			Goal		
7	2	4		1	2
5		6	3	4	5
8	3	1	6	7	8

$$h_1 = 8, \quad h_2 = 18, \quad h^* = 26.$$

Both are safe; neither captures all tile interactions.

Every misplaced tile contributes at least one to Manhattan distance, so $h_2(n) \geq h_1(n)$ for every state.

Dominance and Search Effort

Depth d	Nodes generated			Effective branching factor b^*		
	BFS	A^*+h_1	A^*+h_2	BFS	A^*+h_1	A^*+h_2
12	2,672	279	84	1.80	1.45	1.28
20	91,493	9,905	1,318	1.69	1.50	1.34
28	463,234	202,565	22,055	1.53	1.49	1.36

Dominance

For consistent h_a, h_b , if $h_a(n) \geq h_b(n)$ for every n , then h_a dominates h_b ; A^* expands no more nodes, aside from tie effects.

Effective branching factor

$N + 1 = 1 + b^* + (b^*)^2 + \dots + (b^*)^d$. Values closer to 1 indicate more focused search.

A small reduction in b^* compounds across depth.

Composite and Pattern Heuristics

Combine safe estimates

$$h_{\max}(n) = \max_i h_i(n).$$

If every h_i is admissible, the maximum is admissible and dominates every component.

Why not add blindly?

$h_1 + h_2$ may count the same required work twice. Sums need a cost-partition or disjointness argument.

Precompute exact subproblem costs

1. keep a selected tile pattern,
2. solve every abstract pattern backward from the goal,
3. store each exact abstract cost,
4. look it up during A*.

Pattern-database tradeoff

Stronger lower bounds and cheap lookup during search, paid for with precomputation and memory.

Heuristic Tradeoffs

Heuristic	Guarantees		Guidance	Extra cost	Main limitation
	A	C			
$h_0 = 0$	Yes	Yes	None	Negligible	Reduces A* to UCS
Misplaced tiles	Yes	Yes	Low	Very low	Ignores distance
Manhattan	Yes	Yes	Medium	Low	Ignores interactions and obstacles
$\max_i h_i$	Yes	Yes [†]	$\geq$ each part	Evaluate all components	Evaluation may erase gains
Pattern database	Yes	Usually [‡]	High	Precompute, memory, lookup	Quality depends on abstraction

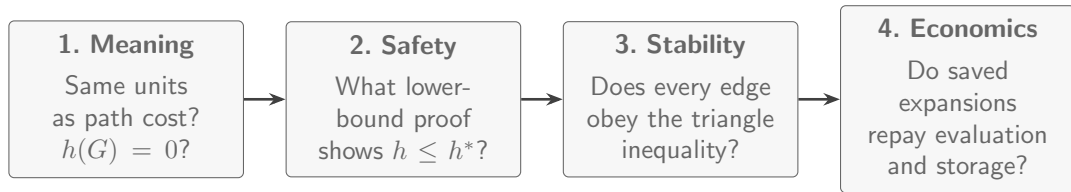
A: admissible; **C**: consistent.

[†] All components must be consistent.

[‡] The abstraction must define a valid lower bound.

Compare **total problem-solving cost**, not node count alone.

Heuristic Design Checklist



Opening question 1

Larger is better only while the value remains safe and is worth computing.

Opening question 2

Overestimation can remove A^* 's lowest-cost guarantee.

Opening question 3

Yes: admissible but inconsistent h can require reopening.

Heuristic-Design Takeaways

Admissibility

A global lower bound protects A*'s solution-cost guarantee.

Consistency

Local edge triangles make f monotone and remove expanded-state reopens.

Construction

Relaxations and abstract subproblems turn provable lower bounds into heuristics.

Quality

Dominance can reduce expansions, but evaluation and memory still count.



Thank You

Any questions?

References

S. J. Russell and P. Norvig. *Artificial Intelligence: A Modern Approach*. Pearson, 4th edition, 2020. URL <https://aima.cs.berkeley.edu/>.